Siddeswararao Botcha

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LinkedIn GitHub Portfolio

Professional Summary

Computer Science graduate skilled in Python, SQL, Data Science, Machine Learning, NLP, and Generative AI, with hands-on internship experience in AI and Data Science. Experienced in data preprocessing, EDA, feature engineering, statistical analysis, and model development using Scikit-learn. Built AI/ML projects using LLMs, RAG, Transformers, NLP, and Flask, seeking entry-level opportunities in Data Science, Machine Learning, or AI Engineering.

Technical Skills

- **Programming Languages:** Python, SQL
- **Python Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn
- **Data Analysis:** Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Data Visualization
- **Statistics:** Descriptive Statistics, Probability, Hypothesis Testing, Correlation, Regression Analysis
- **Machine Learning:** Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, SVM, Naive Bayes, K-Means Clustering, XGBoost, AdaBoost
- **Model Evaluation:** Accuracy, Precision, Recall, F1-Score, Confusion Matrix, MAE, MSE, RMSE, R2 Score
- **Deep Learning:** ANN, CNN, RNN, LSTM, Transformers, Activation Functions
- **NLP & Generative AI:** NLP, LLMs, Prompt Engineering, RAG, Embeddings, Vector Databases, AI Agents
- **Tools & Frameworks:** Jupyter Notebook, VS Code, Git, GitHub, Power BI, Flask, FastAPI, Streamlit

Projects

Employee Salary Growth Prediction *Python, Pandas, NumPy, Scikit-learn, Matplotlib*

- Built a Random Forest regression model to predict employee salary growth using Python and Scikit-learn.
- Performed data preprocessing, categorical encoding using OneHotEncoder, and train-test splitting.
- Evaluated regression models using MAE, MSE, RMSE, and R2 metrics, achieving an R2 score of 85% on the test set.

Customer Segmentation using Machine Learning *Python, Pandas, Scikit-learn, K-Means*

- Developed a K-Means clustering model to group customers based on annual income and spending score.
- Performed data preprocessing, feature selection, and standardization for customer data.
- Analyzed customer segments to recommend personalized marketing and customer retention strategies.

AI-Powered RAG Chatbot *Python, Flask, React, Gemini API, RAG, Git, Render*

- Developed an AI-powered chatbot using React, Flask, and Google Gemini API for real-time conversational responses.
- Built REST APIs with Flask and securely managed API credentials using environment variables.
- Deployed the application using Render and maintained the source code using Git and GitHub.

Professional Experience

BDreamz Global Solutions Chennai, India
Data Science & AI Intern Apr 2025 – Aug 2025

- Developed Python-based data preprocessing pipelines using Pandas, NumPy, and SQL to clean, transform, and prepare structured datasets.
- Performed EDA, statistical analysis, feature engineering, and data validation to identify patterns, handle data quality issues, and improve dataset readiness.
- Trained, validated, and evaluated Scikit-learn regression and classification models using standard performance metrics.
- Explored NLP, LLM, and RAG techniques for AI-powered applications.

Education

Aditya Degree College Jun 2023 – May 2026

B.Sc. Computer Science **CGPA: 8.0 / 10.0**

Certifications

- Python Essentials – Cisco Networking Academy
- Introduction to Machine Learning – Kaggle Learn